

PORTFOLIO CASE STUDY

# Olist E-Commerce Analysis

From nine raw CSVs to business insight with Python, PostgreSQL and Power BI.

DATA → DECISIONS

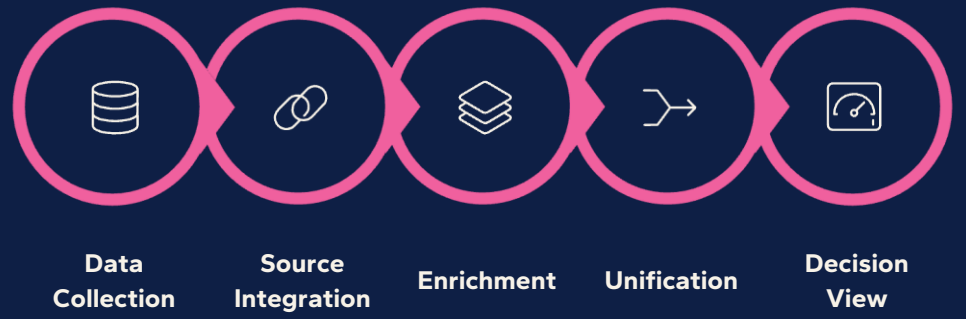
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THE STARTING POINT

# A marketplace story hidden inside nine disconnected tables

Customers, orders, products, sellers, payments, reviews and geography describe the full order lifecycle.

The project goal: turn fragmented marketplace data into decisions a business user can read at a glance.




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ANALYTICAL WORKFLOW

# The workflow moved from raw data to a decision-ready dashboard

1

## Explore

Every table in Python for shape, types, nulls and duplicates.

2

## Clean

Missing values, dates and product attributes standardized.

3

## Query

Nine tables loaded into PostgreSQL; ten business questions answered with SQL.

4

## Visualise

Findings turned into an interactive Power BI dashboard.

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TRUST LAYER

# Data quality work made the analysis trustworthy

| | |
|---|---|
| 01 | Reviews without text were labelled “no review.” |
| 02 | Timestamps were converted to datetime; interpolated values are estimates, not observed events. |
| 03 | Missing product categories used the mode; numeric attributes used the median. |
| 04 | Every table was checked for shape, types, nulls and duplicates. |

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BUSINESS QUESTIONS

# Ten SQL questions translated the dataset into business levers



Revenue and seller performance



Freight and delivery performance



Payment behavior and installments



Customer geography



Review scores and product-category revenue



Monthly orders, deliveries and sales

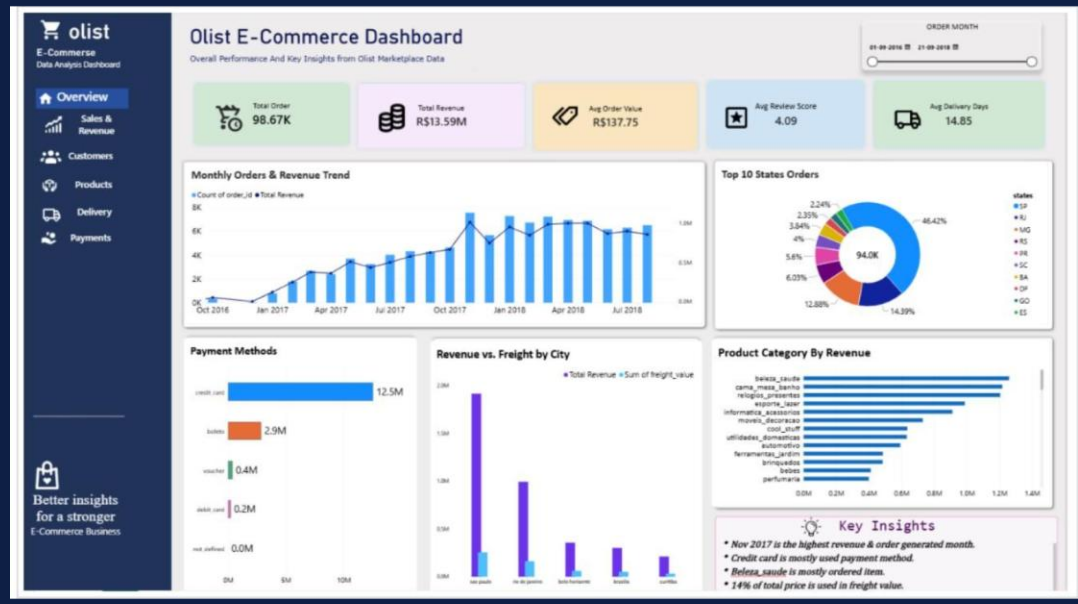
Joins, aggregations, date functions, conditional counts and windowed time views connected the answers.

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DASHBOARD OUTPUT

# The dashboard turns marketplace complexity into five headline measures

| | |
|---|---|
| 98.67K | total orders |
| R$13.59M | total revenue |
| R$137.75 | average order value |
| 4.09 | average review score |
| 14.85 | average delivery time in days |




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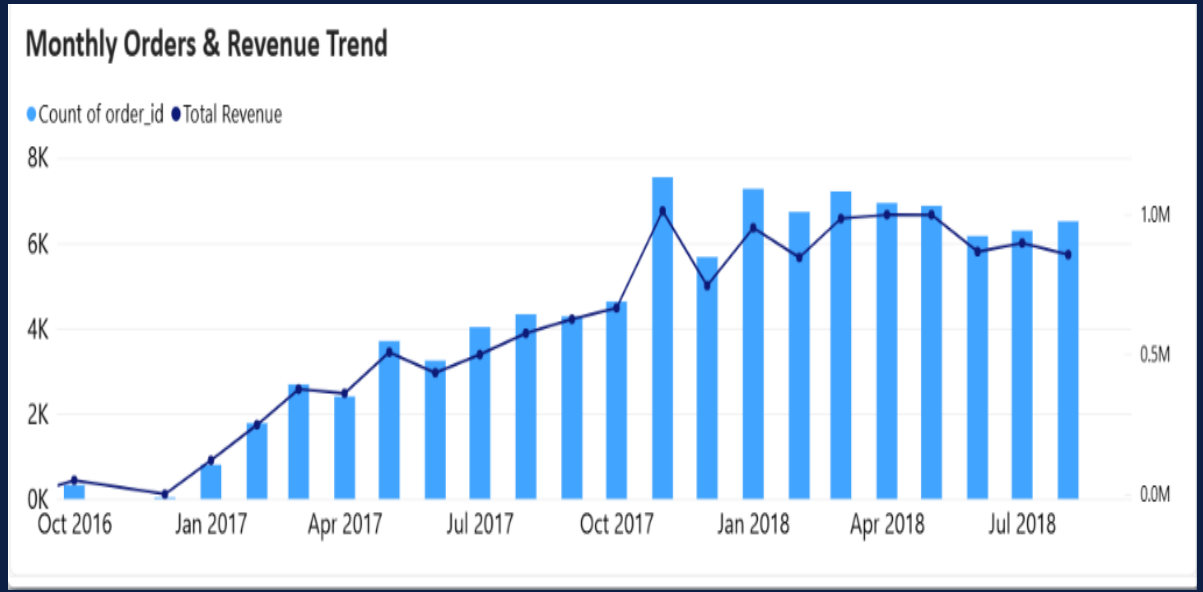
TIME SERIES

## November 2017 was the sales peak

# NOV 2017

November 2017 recorded the highest monthly orders and revenue in the analysed period.

Orders and revenue rose from late 2016. Interpret the 2018 trend alongside dataset coverage and filters.




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WHERE DEMAND COMES FROM

# Demand is concentrated in Brazil’s southeast, and payments are card-led

Share of orders within the top 10 states

| State | Orders |
|---|---|
| São Paulo | 46.42% |
| Rio de Janeiro | 14.39% |
| Minas Gerais | 12.88% |

Together, these three states account for nearly 74% of orders within the top 10 states.

## Payment value

| Payment method | Value |
|---|---|
| Credit card | R$12.5M |
| Boleto | R$2.9M |

Credit card payments lead at R$12.5M, ahead of boleto at R$2.9M.

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CATEGORY AND COST

# Health and beauty leads revenue, while freight remains a visible cost

**14%**

of combined product price and freight value is freight.

Shipping is a meaningful part of customer cost.

## Revenue leaders

1. **beleza_saude** — Top revenue category
2. **cama_mesa_banho**
3. **relogios_presentes**

São Paulo and Rio de Janeiro lead both revenue and freight totals in the city comparison.

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NEXT ANALYTICAL MOVES

# The next analytical moves are about retention, delivery and forecasting

1

## Retention

Study repeat purchases with *customer_unique_id* to measure retention.

2

## Delivery

Link delivery delays to review scores to quantify the customer-experience effect.

3

## Sellers

Expand seller-performance analysis beyond the single top seller.

4

## Forecasting

Add a sales forecast to support planning.

The project demonstrates an end-to-end analytics workflow: clean data, relational analysis and decision-ready storytelling.

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